

Configuration Sensitivity in LLM Evaluation: An Artifact-Based Case Study

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Abstract

Language-model comparisons can depend on the evaluation configuration as well as the model. We investigate how extracted-answer scores and point-estimate rankings differ across recorded prompt, backend, quantization, and decoding configurations. We audit an existing evaluation repository and reanalyze 31 completed runs on the same 800-item slice of ARC-Easy, GSM8K, HellaSwag, and multilingual grade-school mathematics. Three instruction-tuned models have a complete shared greedy-configuration matrix; a fourth model has only a single quantized measurement. We recompute accuracy from integer correctness counts, compare paired outcomes, and report configuration-specific rankings. The largest observed within-model prompt range is 17.375 percentage points, compared with a 2.625-point control gap between the two highest-scoring models. Five-shot prompting reverses the point-estimate ordering of the two highest-scoring models, with a Qwen-minus-Phi difference of 12.250 points. Other shared configurations preserve the control ordering. Recorded greedy-seed correctness vectors are identical, whereas sampled decoding changes item outcomes despite smaller net score movement. These are preliminary, protocol-specific findings: software environments evolved, model revisions were not archived, and the permissive answer extractor has not been independently validated. The artifacts support descriptive configuration-conditional reporting, not isolated causal effects or general leaderboard reliability. We provide a reproducible analysis pipeline and identify the matched-environment and scoring-validation experiments needed to strengthen the evidence.

1 Introduction

Benchmark comparisons inform model selection, but the reported score is produced by an evaluation procedure. Prompt instructions, demonstrations, answer extraction, inference implementations, and decoding defaults can all enter that procedure. Reproducible evaluation therefore requires more than naming a model and benchmark [1]. Prior work already establishes prompt sensitivity and changes in relative model performance [8, 9]; we do not claim to discover this phenomenon.

Our question is narrower: what can an existing configuration-aware evaluation artifact actually establish about score and ranking sensitivity? We analyze the recorded outputs rather than treating dashboard summaries, synthetic tests, or planned runs as empirical observations. This distinction is important because recorded configuration labels do not establish identical effective environments.

The strongest descriptive finding is a change in the ordering of Qwen2.5-3B-Instruct and Phi-3.5-mini-instruct under a five-shot prompt. This accompanies substantial within-model score movement. The study also exposes provenance and measurement limitations, which we report alongside the results. We therefore frame benchmark scores as configuration-conditional measurements in this setting, not as evidence for a universal law or a validated reliability metric.

Our contributions are:

1. An artifact audit distinguishing completed measurements, software demonstrations, and un-run experimental plans, including explicit provenance gaps.
2. A count-based reanalysis of recorded score sensitivity with paired comparisons, separately defined multiplicity families, and pointwise uncertainty intervals.
3. A descriptive ranking analysis over the complete shared matrix, with scripts connecting run artifacts to generated tables and vector figures.

2 Related Work

HELM emphasizes scenario coverage and multiple metrics [5]; BIG-bench broadens the task space [11]. The evaluation-harness literature documents comparability and reproduction problems [1]. Our implementation is not evidence that existing harnesses omit these concerns.

FormatSpread examines meaning-preserving prompt-format variation [9]; multi-prompt evaluation studies absolute and relative model behavior [8]. Our variants are not all meaning-preserving: asking for a final number instead of reasoning changes the requested behavior, and demonstrations change context. We study these operational choices rather than attribute differences to formatting alone. Error-bar methodology distinguishes point estimates from uncertainty [7]. We do not interpret overlapping marginal intervals as equivalence. Recorded backend and quantization bundles are not a controlled systems comparison. These findings do not establish performance on MMLU, BIG-bench, dynamic benchmarks, or public leaderboards.

3 Evaluation Setting

Let M denote models, T tasks, and C evaluation configurations. For frozen items D_t , define

$$S(m, t, c) = |D_t|^{-1} \sum_{i \in D_t} \mathbf{1}\{e_t(g(m, i, c)) = y_i\},$$

where g is generated text, e_t is the task’s answer extractor, and y_i is the gold answer. The score measures extracted-answer agreement, not independently verified latent capability. A full c includes weights/tokenizer revisions, rendered prompts, software, hardware, precision and generation defaults, not merely a registry label.

Rather than only $S(m, t, c_0)$, we examine $\{S(m, t, c) : c \in C_{\text{observed}}\}$. Equal task-slice sizes make pooled accuracy equal to the unweighted mean across tasks. Define $R_c(m)$ by descending pooled accuracy. We report Kendall’s τ_b against control and strict pairwise win fractions

$$\hat{P}(A > B) = \frac{\sum_{c \in C_{AB}} \mathbf{1}\{S(A, c) > S(B, c)\}}{|C_{AB}|}.$$

Here C_{AB} contains observed shared configurations; ties are not wins. These fractions describe our selected matrix, not calibrated probabilities under future configuration choices.

4 Experimental Setup

The evidence boundary is the successful-run registry, not every file in the repository. It contains 31 measurements: eight shared greedy configurations for SmolLM2-1.7B-Instruct, Qwen2.5-3B-Instruct, and Phi-3.5-mini-instruct; three sampled seeds each for Smol and Qwen; and one Qwen2.5-

7B-Instruct INT4 cell. The last model is excluded from shared rankings. The planned Phi sampling arm is missing. No Apertus model appears in this evidence boundary.

Each run contains the same 800 IDs: 200 each from ARC-Easy test, GSM8K test, HellaSwag validation, and MGSM test pooled across English, German and French [2, 3, 10, 13]. The snapshot script shuffles with seed 0 for scored items and seed 1 for exemplar selection. Hub revisions are documented in the data manifest; local slice bytes are hashed by our pipeline. Repeated outputs across configurations are not independent benchmark questions. MGSM translations and GSM8K-related questions create potential dependence even between distinct local IDs.

The declared control uses HF generation, tokenizer chat-template selection, default zero-shot instructions, temperature 0, top- p 1, seed 0, no quantization, maximum 256 new tokens and batch size 1. Variants change prompt ID (concise or five-shot), backend (vLLM), quantization (HF bitsandbytes INT8 or NF4), or greedy seed (1 or 2). Sampling jointly changes temperature to 0.7 and top- p to 0.95, with seeds 0, 1, and 2: a decoding-policy bundle, not an isolated temperature effect. Default math instructions request reasoning; concise requests only an answer. Five-shot uses frozen answer-only demonstrations. The appendix specifies current instruction and exemplar files.

The design is nominally one-factor-at-a-time (OFAT), relative to a declared baseline. Strict OFAT requires all other effective settings to remain fixed. Historical software drift, unrecorded revisions and precision resolution prevent verifying that condition here. We report associations, not isolated causal backend/quantization effects. No completed real factorial matrix is present in the selected registry. OFAT cannot recover interactions among configuration dimensions; heterogeneous model responses do not substitute for an interaction experiment.

Recorded environment. Manifests report Tesla T4 hardware. Precision is recorded as `dtype=auto`, not resolved arithmetic precision. Model and tokenizer revisions: *Version not recorded in the current artifact*. Rendered chat templates, driver versions, and complete dependency locks are unavailable. The audit records multiple Python, Transformers, vLLM and harness version strings; these are per-run metadata, not a single shared environment. Current backend code has different precision-selection paths with and without quantization. We cannot certify byte-identical historical prompts or effective generation defaults across backends.

4.1 Statistics

We re-score every stored generation using the current extractor and verify correctness, predictions, frozen IDs and gold answers against the artifacts. Accuracy uses exact integer counts; rounding occurs only for display. Wilson marginal intervals [12] describe a binomial approximation. Paired differences use task-stratified item bootstrap percentile intervals (4,000 replicates, RNG seed 0). Within each task, paired correctness differences are resampled with replacement, preserving task sizes. This quantifies conditional item-sampling uncertainty, not configuration selection, software drift or language/question-cluster dependence.

We use two-sided exact conditional McNemar tests [6], conditioning on the total number of discordant item pairs. Under the null and independent pairs, wins are binomial with probability one half. No discordant pairs yields $p = 1$. Holm adjustment [4] is applied without intermediate rounding to 27 same-model non-control contrasts; a separate post-hoc family contains eight shared Phi/Qwen comparisons. That pair was selected after observing the top scores, so these tests are exploratory, not confirmatory. Intervals are pointwise, not multiplicity-adjusted; related questions may make item-independent inference optimistic. Non-significance does not establish equivalence.

Table 1: Control extracted-answer accuracy. Wilson intervals are conditional binomial approximations, not equivalence tests.

Model	Correct/items	Accuracy (%)	95% interval
SmolLM2-1.7B	318/800	39.750	[36.42, 43.18]
Qwen2.5-3B	515/800	64.375	[61.00, 67.62]
Phi-3.5-mini	536/800	67.000	[63.67, 70.17]

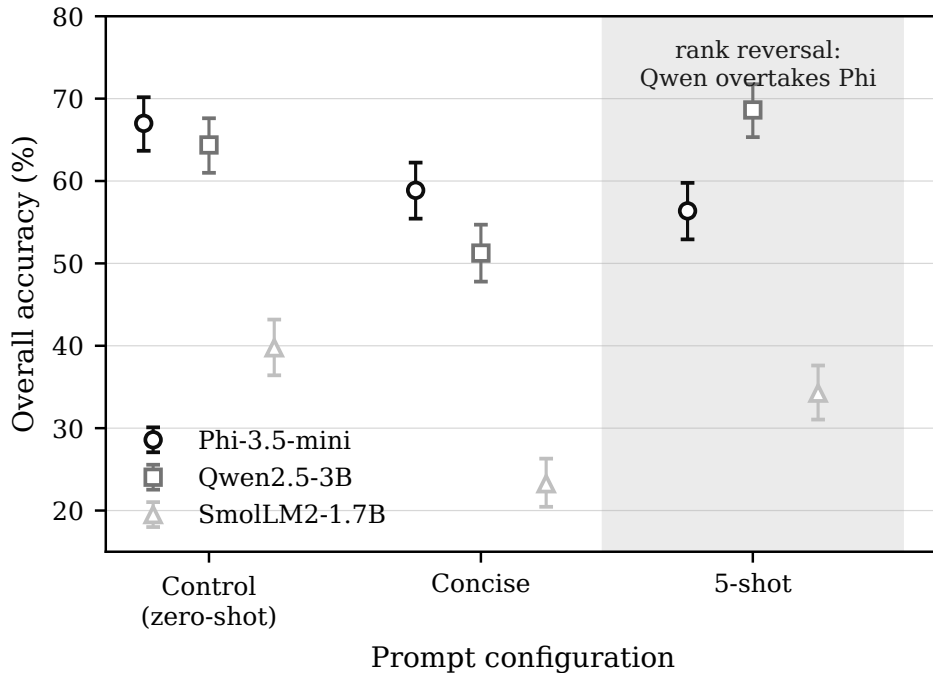


Figure 1: Extracted-answer accuracy under recorded prompt variants. Marginal Wilson intervals use the same frozen items and are correlated across conditions. Five-shot changes the top-two ordering; paired differences appear in Table 3.

5 Results and Sensitivity

The largest observed within-model prompt range is Qwen’s 17.375 percentage points, compared with the 2.625-point Phi-over-Qwen gap under control (Table 1 and Fig. 1). This comparison is descriptive: ranges depend on the variants selected. It is not a variance decomposition or a claim that prompts always dominate evaluation uncertainty. Concise reduces all three models’ pooled scores. Five-shot has heterogeneous effects: Qwen improves, while Smol and Phi decline. These observations involve both instruction and context changes, not a purely formatting effect. Table 2 reports paired effects and uncertainty. No backend or quantization contrast meets a family-wise threshold of 0.05 in this exact-test analysis; this does not establish absence or practical equivalence. The comparisons remain subject to provenance confounding.

Table 2: Variant minus control (percentage points). Task-stratified paired bootstrap intervals; exact McNemar p-values adjusted with Holm over all non-control same-model contrasts, including seeds and sampling. Intervals are pointwise, not simultaneous.

Model	Variant	Difference (pp)	95% interval	Holm p
SmolLM2-1.7B	vllm	+2.250	[0.88, 3.62]	0.0646
SmolLM2-1.7B	5shot	-5.500	[-8.88, -2.25]	0.0356
SmolLM2-1.7B	concise	-16.500	[-20.25, -12.75]	< 0.001
SmolLM2-1.7B	int4	-1.125	[-4.50, 2.25]	1
SmolLM2-1.7B	int8	+2.000	[-0.38, 4.12]	1
Qwen2.5-3B	vllm	+2.375	[0.25, 4.50]	0.726
Qwen2.5-3B	5shot	+4.250	[1.50, 7.00]	0.0709
Qwen2.5-3B	concise	-13.125	[-15.88, -10.38]	< 0.001
Qwen2.5-3B	int4	+1.250	[-1.25, 3.62]	1
Qwen2.5-3B	int8	+0.375	[-2.00, 2.62]	1
Phi-3.5-mini	vllm	+0.125	[-1.50, 1.75]	1
Phi-3.5-mini	5shot	-10.625	[-14.00, -7.38]	< 0.001
Phi-3.5-mini	concise	-8.125	[-11.25, -4.88]	< 0.001
Phi-3.5-mini	int4	+2.875	[0.25, 5.50]	0.726
Phi-3.5-mini	int8	+0.250	[-2.00, 2.38]	1

Table 3: Qwen minus Phi: paired model comparison. Holm correction covers the separate family of shared-configuration comparisons. Positive values favor Qwen. Conditional, post-hoc analysis; no claim of equivalence from non-significance.

Configuration	Difference (pp)	95% interval	Holm p
control	-2.625	[-5.62, 0.38]	0.47
concise	-7.625	[-10.62, -4.62]	< 0.001
5shot	+12.250	[9.00, 15.50]	< 0.001
vllm	-0.375	[-3.38, 2.50]	0.87
seed 1	-2.625	[-5.62, 0.38]	0.47
seed 2	-2.625	[-5.62, 0.38]	0.47
int8	-2.500	[-5.50, 0.62]	0.47
int4	-4.250	[-7.38, -1.25]	0.0559

5.1 Ranking stability

Table 4 shows that only five-shot changes the point-estimate ordering within the shared matrix. It places Qwen ahead of Phi by 12.250 points. Concise instead favors Phi; both comparisons are supported by the conditional paired analysis (Table 3). Control’s Holm-adjusted paired $p = 0.470$ does not establish equivalence. Overlapping Wilson intervals are not a “statistical tie.” Phi has a higher point estimate than Qwen in seven of eight selected configurations; each beats Smol throughout. Repeated greedy-seed columns give control-like outcomes extra weight in this fraction. It is not a probability of outperforming on a future benchmark. Large score changes need not change rankings (concise); the five-shot ordering change answers a different question from score sensitivity. We omit configuration-bootstrap ranking probabilities from central evidence because variants were deliberately selected rather than sampled from a specified population.

Table 4: Ranks ordered Smol/Qwen/Phi, Kendall tau-b against control, and pairwise reversals. Ties would receive shared ranks; none occur in these columns.

Configuration	Ranks	τ_b	Reversals
control	3/2/1	1.000	0
concise	3/2/1	1.000	0
5shot	3/1/2	0.333	1
vllm	3/2/1	1.000	0
seed 1	3/2/1	1.000	0
seed 2	3/2/1	1.000	0
int8	3/2/1	1.000	0
int4	3/2/1	1.000	0

5.2 Decoding, failures, and efficiency

The recorded greedy-seed correctness vectors match control exactly. This establishes correctness invariance in these runs, not identical text, tokens, or general decoding determinism. Sampled decoding changes item correctness while producing smaller net shifts; the missing Phi sampling runs preclude a complete sampled ranking comparison. Sampling temperature and top- p both differ from control.

The extractor assigns multiple-choice letters or numeric answers from generated text. A wrong extracted answer may reflect reasoning, copied demonstrations, misleading intermediate text, or an extractor error. Rare explicit unparseable outputs do not validate extraction, and we do not infer causal failure mechanisms from those counts. All recorded scored items, including malformed or empty outputs if present, remain in accuracy denominators. Successful-run selection cannot establish the absence of timeouts or failures in unarchived attempts. The artifact-backed failure table in the appendix distinguishes these scopes.

Timing is ancillary. Recorded vLLM zeros are unavailable measurements, not zero latency. HF measurements are specific to their runtime; without matched effective settings and validated instrumentation we make no comparative quality-efficiency claim. Memory, monetary cost, and energy trade-offs are not established by the paper matrix.

6 Discussion

The observed score movements and ranking change are consistent with treating a benchmark score as a configuration-conditional measurement here. This is an empirical framing, not a universal property proven here. Prior prompt-sensitivity work already motivates this perspective; our contribution is a transparent artifact reanalysis rather than methodological priority. A score range and a ranking reversal answer different questions. Both need task, configuration, provenance and measurement assumptions.

7 Limitations and Missing Experiments

Measurement validity. Generative letter/number extraction differs from log-likelihood multiple-choice evaluation. No leaderboard comparison is warranted. The permissive parser has not been independently adjudicated; archived correctness is internally reproducible but may not reflect the intended final answer. A stratified blinded output review and prespecified parser

comparisons are necessary before stronger capability claims. Contamination is unmeasured and may interact with prompting; keeping items fixed does not remove it.

Provenance and control. Unpinned model/tokenizer revisions, unarchived rendered prompts and changing software prevent exact historical inference reproduction. Declared `auto` precision does not prove matched precision. Settings hashes omit relevant source, data and environment context. Recorded Git IDs identify intended versions but some corresponding objects are absent locally. We cannot reconstruct missing information retrospectively or certify historical hardware from manifest strings alone. Headline contrasts require reruns with immutable revisions, prompt/token archival, resolved precision, complete environments, and fixed scoring.

Scope and dependence. The shared ranking covers small instruction-tuned models, selected configurations, small task slices and one recorded accelerator family. MGSM translations and related GSM8K questions are not independent semantic problems. Item-wise intervals and tests are conditional approximations; recovering source-question groups and cluster-aware inference is a priority. Post-hoc model-pair selection, limited prompt sampling and repeated greedy-seed columns restrict interpretation. There is no claim about frontier models, other languages, safety or general benchmark validity. Missing Phi sampled cells must be completed before reporting full sampled rankings.

Interactions and proposed diagnostics. The repository contains a future prompt $\times$ runtime $\times$ seed plan, not completed factorial evidence in the paper registry. No interaction effect is estimated. The Evaluation Reliability Score combines interval separation, bootstrap ranking stability, configuration stability and optional seed stability using provisional weights. It is exploratory tooling, not a calibrated metric; pooling repeated questions and reweighting selected configurations can make it misleading. We omit its numerical composite and do not treat synthetic software tests as empirical validation.

8 Reproducibility

Evidence comes from `results/registry_paper.jsonl` and its raw run JSONs. Analysis verifies identities, labels, rescoring and configuration-hash consistency before computing exact-count summaries. `scripts/statistical_analysis.py` supplies the descriptive figure matrix; `scripts/audited_analysis.py` supplies exact conditional tests, task-stratified bootstrap intervals and tables. Legacy exploratory test implementations are not mixed with the manuscript’s exact-test results.

From the repository root, `make analyze` regenerates and checks assets without inference; `make paper` additionally compiles LaTeX. Select the interpreter with `PYTHON`. Existing NumPy, Matplotlib and PyYAML dependencies are used; Tectonic or PDFLaTeX/BibTeX is required for compilation. Source and asset hashes, per-run settings, package strings and hardware metadata are retained in the analysis directory. Analysis reproduction is feasible; exact historical inference is not established.

The factorial runner supports a dry-run plan separate from paper reproduction. Execution requires model downloads and a compatible GPU environment; its unpinned defaults need review before confirmatory runs. No time or monetary estimate is claimed without measurement. Automated checks cover generated assets and references, not scientific validity or arbitrary numerical prose; a claim audit and human review remain necessary.

9 Ethical and Licensing Considerations

We make no safety, privacy, toxicity, legal-compliance or deployment claims. Contamination and memorization are unmeasured. Dataset/model licenses need review against primary distribution sources before redistributing snapshots or generations; attribution notes are not legal certification. No new private data were collected for this reanalysis. Author identity, affiliation and submission license require author confirmation; this draft has not been submitted or peer reviewed.

10 Conclusion

In the recorded matrix, prompt configurations coincide with substantial extracted-answer score movement and a top-two ordering change. Other shared configurations preserve the ordering. Paired analysis makes these observations inspectable, but scoring and provenance concerns preclude stronger causal or general reliability conclusions. Configuration-conditional reporting is a useful framing here. Matched-environment confirmation and output adjudication, rather than a larger narrative or unvalidated composite, are the next steps.

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A Artifact and Protocol Details

Evidence boundary. Only successful rows of `results/registry_paper.jsonl` enter this analysis. Smoke runs, synthetic factorial fixtures, dashboards and archived copies are not additional experiments. The machine-readable audit records file hashes, notebook cells, run metadata and availability of recorded Git objects. Manifest values describe what was recorded, not independent authentication of execution.

Exact protocol sources. Prompt instructions are in `configs/prompts/default.yaml`, `configs/prompts/concise.yaml` and `configs/prompts/5shot.yaml`. Task exemplars and scored items are frozen in `data/official/fewshot.jsonl` and `data/official/eval_set.jsonl`. The default multiple-choice instruction is “Reply with the letter of the correct answer (A, B, C, or D).” Default math asks “Reason step by step. Put the final numeric answer on the last line.” No system instruction is specified in the default YAML. Current prompt assembly is in `src/apertus_eval_prep/prompting.py`. These sources do not reconstruct missing historical tokenizer-template bytes.

Extraction. `src/apertus_eval_prep/scoring.py` first checks a case-insensitive A–D prefix for multiple-choice outputs, otherwise the last standalone A–D letter. Prefix matching can accept the beginning of an ordinary word. Math extracts the last regex-matched number after removing commas and normalizing numeric representation. Thus parseability alone is not semantic correctness. These rules are included to make limitations inspectable, not endorsed as a validated scoring protocol.

Table 5: Post-hoc output categories across all successful paper runs, including sampled and single-cell runs. Denominator counts repeated outputs, not independent questions. Runtime failures/timeouts outside archived successful runs are unknown, not zero.

Model	Outputs	Correct	Incorrect	Unparseable	Empty
SmolLM2-1.7B	8800	3296	5502	2	0
Qwen2.5-3B	8800	5619	3181	0	0
Qwen2.5-7B	800	543	257	0	0
Phi-3.5-mini	6400	4164	2235	1	0

Table 6: Control task-level extracted-answer accuracy; the full task breakdown for every run is exported in run details.

Model	Task	Correct	Items	Accuracy (%)
SmolLM2-1.7B	arc_easy	144	200	72.00
SmolLM2-1.7B	gsm8k	64	200	32.00
SmolLM2-1.7B	hellaswag	66	200	33.00
SmolLM2-1.7B	mgsm	44	200	22.00
Qwen2.5-3B	arc_easy	192	200	96.00
Qwen2.5-3B	gsm8k	85	200	42.50
Qwen2.5-3B	hellaswag	154	200	77.00
Qwen2.5-3B	mgsm	84	200	42.00
Phi-3.5-mini	arc_easy	197	200	98.50
Phi-3.5-mini	gsm8k	95	200	47.50
Phi-3.5-mini	hellaswag	160	200	80.00
Phi-3.5-mini	mgsm	84	200	42.00

Data revisions and environments. Full Hub revision strings and declared licenses are in `data/official/SOURCES.md`; local source SHA-256 values are recorded in `analysis/audited_statistics.json`. Per-run package/hardware/settings records are exported with that analysis. The following generated tables report artifact counts and descriptive failures without interpreting low malformed-output frequency as extractor validity.

Remaining work. The unrun crossed design is `configs/experiments/t4_factorial.yaml`, with runner `scripts/run_t4_research.py`. It does not repair historical confounding. Confirmatory runs must pin revisions, hold environments fixed, archive effective prompts and defaults, and log every failed attempt. Output adjudication and recovering semantic question clusters precede stronger inferential claims. More models/tasks and randomized prompt/exemplar sets are required for broader generalization; they are not completed here.